

Business Analytics Case Study Report

Customer Churn Analysis Using Python

Prepared by: Azeemah Sullayman

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Executive Summary

Customer churn has a direct impact on revenue, customer lifetime value, and business growth. This case study analyzes the Telco Customer Churn dataset to identify the major factors influencing customer attrition and provides actionable recommendations that can improve customer retention and support data-driven decision-making.

Business Problem

The company is experiencing customer churn but lacks a clear understanding of the characteristics and behaviours associated with customers who leave. The objective is to identify the primary drivers of churn and recommend practical retention strategies.

Project Objectives

- Understand customer churn patterns.
- Identify the major drivers of customer churn.
- Generate business insights from the dataset.
- Provide actionable recommendations for improving retention.

Dataset Overview

The Telco Customer Churn dataset contains customer demographics, subscribed services, contract information, payment methods, monthly charges, total charges, tenure, and churn status. The dataset was analyzed using Python in Jupyter Notebook.

Data Preparation

- Inspected dataset structure and data types.
- Checked for missing values and duplicates.
- Converted the TotalCharges column to numeric format.
- Detected and removed 11 hidden missing records.
- Exported a cleaned dataset for dashboard development in Power BI.

Exploratory Data Analysis Summary

Exploratory analysis focused on identifying the key factors associated with customer churn through eleven business-focused visualizations covering customer demographics, account information, subscribed services, billing behaviour, and churn distribution.

Key Business Findings

- Approximately one in four customers churned.
- Month-to-month contracts experienced the highest churn.
- Customers with shorter tenure were significantly more likely to leave.
- Higher monthly charges were associated with increased churn.
- Customers without technical support exhibited higher churn.
- Internet service type and payment method influenced churn patterns.

Business Recommendations

1. Promote long-term contracts through loyalty incentives.
2. Strengthen onboarding for new customers.
3. Review pricing strategies for customers with high monthly charges.
4. Bundle technical support with internet plans.
5. Use churn indicators to proactively target high-risk customers.

Conclusion

The analysis identified contract type, tenure, monthly charges, technical support, and internet service as the strongest drivers of customer churn. Implementing targeted retention strategies based on these findings can improve customer satisfaction, reduce attrition, and support sustainable business growth.